

Econometrics and Econometricians at Texas A&M

Manu Navjeevan

TEXAS A&M UNIVERSITY

February, 2026

Table of Contents

What is Econometrics?

The Landscape of Econometrics

Econometricians at Texas A&M

What is Econometrics?

Econometrics sits at the intersection of **economics**, **statistics**, and **mathematics**.

- Core question: *How do we learn about causal relationships and economic structure from data?*
- Econometrics is shaped by the realities of economic data:
 - Data is **non-experimental** — we rarely get to run the experiment we want
 - Agents are **strategic** — they respond to incentives and each other
 - We care about specific **causal** and **structural** parameters
- Econometricians develop the **methods and theory** that empirical researchers use

Statistics: General-purpose inference — estimation, testing, prediction — often under well-controlled settings

Econometrics: Inference under constraints imposed by economic reality

- Endogeneity, selection, simultaneity, strategic behavior
- The data-generating process is shaped by economic agents making choices

Example:

- Often times we may be interested in a conditional expectation: $\mathbb{E}[Y \mid X]$
- Statistics asks: *How can we estimate $\mathbb{E}[Y \mid X]$?*
- Econometrics asks: *When does $\mathbb{E}[Y \mid X]$ have a **causal** interpretation?*

- **Applied economists** (labor, IO, trade, macro, . . .) *use* econometric tools to answer substantive questions
- **Econometricians** *develop* those tools — and ask:
 - When do they work?
 - When do they fail?
 - How can we do better?
- Econometrics is its own research field with its own:
 - Journals: *Econometrica*, *Journal of Econometrics*, *Econometric Theory*, . . .
 - Questions, community, and intellectual identity

Table of Contents

What is Econometrics?

The Landscape of Econometrics

Econometricians at Texas A&M

Identification vs. Estimation

Two fundamental questions in econometrics:

Identification: What *can* we learn, even with infinite data?

- Is the parameter of interest pinned down by the model and data structure?

Estimation: *How* do we best learn it from a finite sample?

- Rate of convergence, efficiency, robustness

These are complementary but distinct — a parameter can be identified but hard to estimate, or easy to “estimate” but not identified.

Example (IV): $Y = X'\beta + \varepsilon$, with instrument Z

- β is **identified** iff $\mathbb{E}[Z\varepsilon] = 0$ and $\mathbb{E}[ZX']$ has full rank
- **Estimation:** 2SLS, LIML, ... — each with different finite-sample properties

How Econometrics Connects to Empirical Research

Econometric theory underpins every major empirical strategy:

- **Design-based / reduced-form:** IV, RDD, DID
 - Credible identification from quasi-experiments
- **Structural estimation:** Specify and estimate economic models
 - Auctions, dynamic discrete choice, games of entry
- **Machine learning / high-dimensional methods:**
 - Prediction, variable selection, heterogeneous treatment effects

Each requires econometric theory to understand *what* is being estimated and *why*.

What do econometricians work on?

- Developing new **estimators and tests**
- Characterizing what is **identified** under different assumptions
- Understanding **finite-sample and asymptotic** properties of procedures
- Bridging theory and practice — making methods **usable** for applied researchers

Table of Contents

What is Econometrics?

The Landscape of Econometrics

Econometricians at Texas A&M

Endowed Chair, Department of Economics

- One of the leading researchers in **nonparametric and semiparametric econometrics**
- Co-author of *Nonparametric Econometrics: Theory and Practice* (Li and Racine, 2007) — a standard reference in the field
- Research focus: estimation when we don't want to impose restrictive functional form assumptions
- Key contributions to kernel estimation, bandwidth selection, and nonparametric testing

Nonparametric Estimation — Why It Matters

Parametric models impose structure: $\mathbb{E}[Y | X] = X'\beta$

Nonparametric: Let the data speak — estimate $\mathbb{E}[Y | X] = m(X)$ without specifying m

Kernel estimator:

$$\hat{m}(x) = \frac{\sum_{i=1}^n K_h(X_i - x) Y_i}{\sum_{i=1}^n K_h(X_i - x)}$$

where $K_h(\cdot) = K(\cdot/h)/h$ is a kernel function with bandwidth h

Applications:

- Estimating heterogeneous treatment effects: $\mathbb{E}[Y(1) - Y(0) | X = x]$
- Flexible first-stage estimation in semiparametric models
- Specification testing: is a parametric model adequate?

Associate Professor, Department of Economics

- Research spans **empirical industrial organization** and **measurement error econometrics**
- Uses econometric tools to study strategic behavior:
 - Auctions, contracting, and market design
- Bridges **structural IO** and **econometric theory**
- Develops methods for identification and estimation when data is measured with error or agents deviate from standard behavioral assumptions

(An, 2017)

Standard auction analysis: Bidders play Nash equilibrium

- But do they? What if bidders have **incorrect beliefs** about competitors?

The identification problem changes fundamentally:

- Observed bids reflect true values filtered through (possibly wrong) beliefs
- This is analogous to a **measurement error** problem

Key contributions:

- Shows how to **identify** the distribution of bidder values and belief structures from bid data alone
- Develops estimation methods robust to non-equilibrium behavior

Why it matters: Understanding real auction behavior informs procurement policy, spectrum auctions, and market design

Assistant Professor, Department of Economics

- Research area: **statistical decision theory**
- Core question: How should we **make decisions under uncertainty**, and how do we **evaluate decision procedures**?
- Connects foundational statistical theory to practical econometric problems:
 - Treatment assignment, policy learning, empirical welfare maximization

Setup: We want to estimate the optimal decision/policy

- E.g., optimal treatment assignment: who should receive treatment?

The problem with consistency:

- Easy to construct **consistent** estimators of optimal decisions
- But consistency is a *weak* requirement — throwing away half the data still gives consistency

Key question: How do we **rank** decision procedures? What makes one “better” than another?

Contribution:

- Develops an asymptotic framework providing a **sharper notion of optimality** for decision procedures
- Connects to a broad literature on treatment choice, policy learning, and empirical welfare maximization

Assistant Professor, Department of Economics

- Research interests: **identification, instrumental variables, high-dimensional inference**
- Work connects econometric theory with modern statistical tools
- Central theme: understanding *what we can learn* from commonly used empirical strategies, and developing methods for when standard assumptions fail

(Navjeevan et al., 2024)

Imbens and Angrist (1994): Under **monotonicity**, IV identifies the Local Average Treatment Effect (LATE)

Specific to Binary Model — what do we do when there are multiple treatments or continuous instruments?

- Researchers have proposed a variety of conditions.
- In a general models, what exactly is identified?

Key contributions:

- In arbitrary models, characterizes which causal parameters are identified
- Provides a unifying framework for thinking about different models
- For identified parameters, analysis suggests simple estimators that are consistent and asymptotically normal

Application: Moving to Opportunity experiment

Four faculty spanning a range of econometric research:

- **Qi Li** — Nonparametric & semiparametric estimation
- **Yonghong An** — Industrial organization & measurement error
- **Han Xu** — Statistical decision theory
- **Manu Navjeevan** — Identification & instrumental variables

A group with range — from foundational theory to applied structural work — and opportunities for students interested in econometric theory and its applications.

References

- ¹Y. An, “Identification of first-price auctions with non-equilibrium beliefs: a measurement error approach”, *Journal of Econometrics* **200**, 326–343 (2017).
- ²G. W. Imbens and J. D. Angrist, “Identification and estimation of local average treatment effects”, *Econometrica* **62**, 467–475 (1994).
- ³Q. Li and J. S. Racine, *Nonparametric econometrics: theory and practice*, (Princeton University Press, 2007).
- ⁴M. Navjeevan, A. Santos, and R. Pinto, “Identification in instrumental variable models with misspecified Monotonicity”, *Revise and Resubmit, Econometrica*, 2024.
- ⁵H. Xu, “Characterizing optimal statistical procedures”, *Job Market Paper*, 2024.